onboarding HTTP/TLS/JSON

Interface Design Description

Service ID: *"onboarding"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `onboarding` service of the Onboarding system.

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `onboarding` service interface of the Onboarding system, which provides the Arrowhead onboarding procedure: a device holding only an approved bootstrap certificate obtains its onboarding, device and system certificates and the identities of its systems.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `onboarding` service interface

This document provides the Interface Design Description IDD to the *onboarding – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `onboarding` service interface in detail. Both in terms of its operations (Section 3) and its information model (Section 4).

2.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `onboarding`, port 8435 and base path `/onboarding`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`onboard-with-name`	POST	`/certificate/name`
`onboard-with-csr`	POST	`/certificate/csr`
`onboard-device`	POST	`/device`
`onboard-system`	POST	`/system`
`offboard-device`	POST	`/device/offboard`

3 Service Operations

The interfaces of the `onboarding` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `onboarding` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

3.1 POST /onboarding/certificate/name

Operation: onboard-with-name
Input: NameRequest
Output: Response

Step 1 without CSR: Onboarding generates a key pair and returns the onboarding certificate with its private key (once).

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/certificate/name HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "commonName": "Device1.TestCloud.Company.arrowhead.eu"
}
```

Listing 1: An onboard-with-name invocation.

```
{
  "certificateChain": [
    "string"
  ],
  "onboardingCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
```

```
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "nextStep": "onboard-device"
}
```


Listing 2: An onboard-with-name response.

3.2 POST /onboarding/certificate/csr

Operation: onboard-with-csr
Input: CsrRequest
Output: Response

Step 1: signs the device's onboarding certificate (`on`) from its CSR; returns the chain and the registry descriptors. A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/certificate/csr HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

```

```
{
  "certificateSigningRequest": "<Base64 DER PKCS#10>"
}
```

Listing 3: An onboard-with-csr invocation.

```
{
  "certificateChain": [
    "string"
  ],
  "onboardingCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
```

```
                "path": "/read"
              }
            }
          }
        }
      }
    ],
    "createdAt": "2026-10-22T12:00:00Z",
    "updatedAt": "2026-10-22T12:00:00Z"
  }
 ],
 "nextStep": "onboard-device"
}
```

Listing 4: An onboard-with-csr response.

3.3 POST /onboarding/device

Operation: onboard-device
Input: DeviceRequest
Output: DeviceResponse

Step 2, with the onboarding certificate: signs the device certificate (`de`) and registers the device in the ServiceRegistry.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/device HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "certificateSigningRequest": "<Base64 DER PKCS#10>",
  "addresses": [
    "string"
  ],
  "metadata": {
    "unit": "celsius"
  }
}
```

Listing 5: An onboard-device invocation.

```
{
  "deviceName": "Raspberry1",
  "certificateChain": [
    "string"
  ],
  "deviceCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "approvedSystems": [
    "string"
  ],
  "nextStep": "onboard-device"
}
```

Listing 6: An onboard-device response.

3.4 POST /onboarding/system

Operation: onboard-system
Input: SystemRequest
Output: SystemResponse

Step 3, with the device certificate: signs the system certificate (`sy`) of an approved system and creates its Authentication identity; returns the identity password once.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/system HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "systemName": "TemperatureSensor",
  "certificateSigningRequest": "<Base64 DER PKCS#10>"
}
```

Listing 7: An onboard-system invocation.

```
{
  "systemName": "TemperatureSensor",
  "deviceName": "Raspberry1",
  "certificateChain": [
    "string"
  ],
  "systemCertificate": {
    "certificate": "<Base64 DER>",
    "certificateFormat": "X.509",
    "certificateType": "SYSTEM",
    "publicKey": "string",
    "privateKey": "string",
    "expiresAt": "2026-10-22T12:00:00Z"
  },
  "authenticationMethod": "PASSWORD",
  "credentials": {
    "password": "<password>"
  },
  "services": [
    {
      "instanceId": "TemperatureSensor|temperatureReading|1.0.0",
      "provider": {
        "name": "TemperatureSensor",
        "metadata": {
          "location": "hall-1"
        },
        "version": "1.0.0",
        "addresses": [
          {}
        ],
        "device": {
          "name": "Raspberry1",
          "metadata": {
            "location": "hall-1"
          },
          "addresses": [
            {}
          ],
          "createdAt": "2026-10-22T12:00:00Z",
          "updatedAt": "2026-10-22T12:00:00Z"
        },
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
```

```
      "serviceDefinition": {
        "name": "temperatureReading",
        "createdAt": "2026-10-22T12:00:00Z",
        "updatedAt": "2026-10-22T12:00:00Z"
      },
      "version": "1.0.0",
      "expiresAt": "2026-10-22T12:00:00Z",
      "metadata": {
        "unit": "celsius"
      },
      "interfaces": [
        {
          "templateName": "generic_https",
          "protocol": "https",
          "policy": "CERT_AUTH",
          "properties": {
            "accessAddresses": [
              "192.168.0.10"
            ],
            "accessPort": 8080,
            "basePath": "/temperature",
            "operations": {
              "read": {
                "method": "GET",
                "path": "/read"
              }
            }
          }
        }
      ],
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z"
    }
  ],
  "nextStep": "onboard-device"
}
```


Listing 8: An onboard-system response.

3.5 POST /onboarding/device/offboard

Operation: offboard-device
Output: OffboardingReport

With the device certificate: removes the device's systems, identities and registry entries, revokes the certificates and blacklists the systems.

A successful call returns `200 OK` with the output in the body.

```
POST /onboarding/device/offboard HTTP/1.1
Host: onboarding:8435
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 9: An offboard-device invocation.


```
{
  "deviceName": "Raspberry1",
  "completed": [
    "string"
  ],
  "failed": [
    "string"
  ]
}
```

Listing 10: An offboard-device response.

4 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 4.17, which are used to represent things like hashes, identifiers and texts.

4.1 struct Address

A network address with its type.

Object Field	Value Type	Description
"type"	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
"address"	String	One network address (host name, IPv4, IPv6 or MAC address).

4.2 struct CertificateResult

An issued certificate.

Object Field	Value Type	Description
"certificate"	String	The certificate, Base64 DER.
"certificateFormat"	String	Encoding of the certificate, e.g. `X.509`.
"certificateType"	String	Certificate profile: `ON_BOARDING`, `DEVICE` or `SYSTEM`.
"publicKey"	String	Public key, Base64 X.509 SubjectPublicKey Info.
"privateKey"	String	Generated private key (only without a CSR, returned once).
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.

4.3 struct CsrRequest

Onboarding with the device's own key.

Object Field	Value Type	Description
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.

4.4 struct DeviceRequest

Request of the device certificate.

Object Field	Value Type	Description
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.
"addresses"	Array<String>	Network addresses of the device for the ServiceRegistry.
"metadata"	Object	Device metadata for the ServiceRegistry.

4.5 struct DeviceResponse

The device certificate and the next step.

Object Field	Value Type	Description
"deviceName"	String	The onboarded device.
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"deviceCertificate"	CertificateResult	The issued device certificate (`de`).
"approvedSystems"	Array<String>	Systems the operator approved for the device.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

4.6 struct DeviceResponseDTO

A registered device.

Object Field	Value Type	Description
"name"	String	Device name.
"metadata"	Object	Free-form metadata (JSON object).
"addresses"	Array<Address>	Network addresses with their types.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.7 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

4.8 struct NameRequest

Onboarding with a generated key for a name.

Object Field	Value Type	Description
"commonName"	String	Common name of the requested certificate, `<Name>.<Cloud>.<Organization>.arrowhead.eu`.

4.9 struct OffboardingReport

Outcome of an off-boarding.

Object Field	Value Type	Description
"deviceName"	String	The off-boarded device.
"completed"	Array<String>	Off-boarding steps that succeeded.
"failed"	Array<String>	Off-boarding steps that failed.

4.10 struct Response

Result of the first onboarding step.

Object Field	Value Type	Description
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"onboardingCertificate"	CertificateResult	The issued onboarding certificate (`on`).
"services"	Array<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

4.11 struct ServiceDefinitionResponse

A service definition.

Object Field	Value Type	Description
"name"	String	Service definition name.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.12 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Object Field	Value Type	Description
"templateName"	String	Name of the interface template, snake_case (e.g. `generic_https`).
"protocol"	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
"policy"	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
"properties"	Object	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.13 struct ServiceInstanceResponse

A registered service instance.

Object Field	Value Type	Description
"instanceId"	String	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
"provider"	SystemResponseDTO	The provider system.
"serviceDefinition"	ServiceDefinitionResponse	The service definition.
"version"	Version	Version of the service instance.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"metadata"	Object	Free-form metadata (JSON object).
"interfaces"	Array<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.14 struct SystemRequest

Request of a system certificate.

Object Field	Value Type	Description
"systemName"	String	System to onboard; must be approved for the device.
"certificateSigning Request"	String	PKCS\#10 certificate signing request, Base64 DER.

4.15 struct SystemResponse

The system certificate, identity and the next step.

Object Field	Value Type	Description
"systemName"	String	The onboarded system.
"deviceName"	String	Name of the device hosting the system.
"certificateChain"	Array<String>	Issued certificate first, then the issuing CA (Base64 DER).
"systemCertificate"	CertificateResult	The issued system certificate (`sy`).
"authenticationMethod"	String	Authentication method of the identity, `PASSWORD`.
"credentials"	Object<String>	Identity password of the new Authentication identity (returned once).
"services"	Array<ServiceInstanceResponse>	Service instances of the core services the caller needs next.
"nextStep"	String	The operation the caller performs next in the onboarding procedure.

4.16 struct SystemResponseDTO

A registered system.

Object Field	Value Type	Description
"name"	String	System name.
"metadata"	Object	Free-form metadata (JSON object).
"version"	Version	Version of the system.
"addresses"	Array<Address>	Network addresses with their types.
"device"	DeviceResponseDTO	The device hosting the system, if registered.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.

4.17 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.
Version	JSON String, semantic version `major.minor.patch`.

5 Go and Rust implementation variants

5.1 Go implementation

Primary system folder: `Eclipse-Arrowhead-go/ah5-Onboarding-go`.

The system owns its Go module, launcher and build configuration. Use the per-system `Makefile` and `go.mod`; the containing Go tree is the build source, not a historical Arrowhead copy. Ordinary native roles use independent identities and JSON snapshot state rather than the Java database. Do not assume Java SQL migration, clustering or identical transaction/retry semantics. The registry has its own configuration and optional native AMQP monitor; other roles must be checked against their own registrations.

Listening port. Built-in default 127.0.0.1:8435 (Arrowhead default). The shipped config.json, compose.yaml and Dockerfile use the same port on all interfaces ("listen": "0.0.0.0:8435"; 8540–8555 layout before 2026-11-09). Override with "listen" in the configuration file, SERVER_PORT=N or -server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Fingerprint/system approval, on/de/sy certificate stages, registry/identity provisioning and offboarding. Complete renewal, crash recovery, withdrawal handling and prerequisite discovery; partial compensating saga, not a distributed transaction. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-go/ah5-Onboarding-go/documentation/Onboarding`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

5.2 Rust implementation

Primary system folder: `Eclipse-Arrowhead-rust/ah5-Onboarding-rust`.

The per-system entry point is declared as a binary of `arrowhead-support-rust` in `ah5-support-runtime-rust/Cargo.toml`. Build from the primary Rust workspace with the applicable package/binary selection. Documentation links can resolve into the shared core/support catalogue. Independent process identity and state remain required.

The ordinary native profile uses Tokio/Axum and rustls with peer-identity checks and independent locked JSON state. Network effects are not distributed database transactions. The precise trust, CRL reload and persistence guarantees are those of the native profile, not the Java implementation.

Listening port. Default 127.0.0.1:8435 (Arrowhead default, role onboarding). Override with "listen": "host:port" in the configuration, SERVER_PORT=N or –server.port N; the port overrides keep the configured host.

Native behavior and outstanding work. Enrollment approval administration and unique fingerprint/system assignment. **Bootstrap/on/de TLS profiles and distributed provisioning/renewal/offboarding absent**; calls return501. These are the boundaries recorded by the native compatibility document; later implementation changes still require direct qualification. Native design/route evidence: `Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/Onboarding`.

For this Java interface, select only a native transport and operation explicitly implemented and advertised by that port. A matching method/path does not establish DTO, error, authorization, retry or persistence equivalence. Java MQTT/AMQP bindings are not automatically provided by a Go or Rust role. Differential contract and security checks remain necessary before substitution.

6 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “onboarding – Service Description (Onboarding),” Eclipse Arrowhead, SD 5.3.0, 2026, onboarding-v5.3.0-onboarding-SD.pdf.

7 Revision History

7.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

7.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	